

Ostad

Software Quality Assurance Final Assessment
UI and API Automation Testing

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Course: SQA: Manual & Automation Testing — Batch 20

Platform: Ostad

Introduction and objectives

Software Quality Assurance uses systematic testing to assess whether software behaves as expected and to make defects observable. UI automation validates user-visible workflows through a browser, while API automation validates request/response contracts. Objectives:

(1) automate the three specified SauceDemo scenarios;

(2) author API tests in Postman for Q4–Q8;

(3) assert response codes for every API request;

(4) support individual and sequential execution;

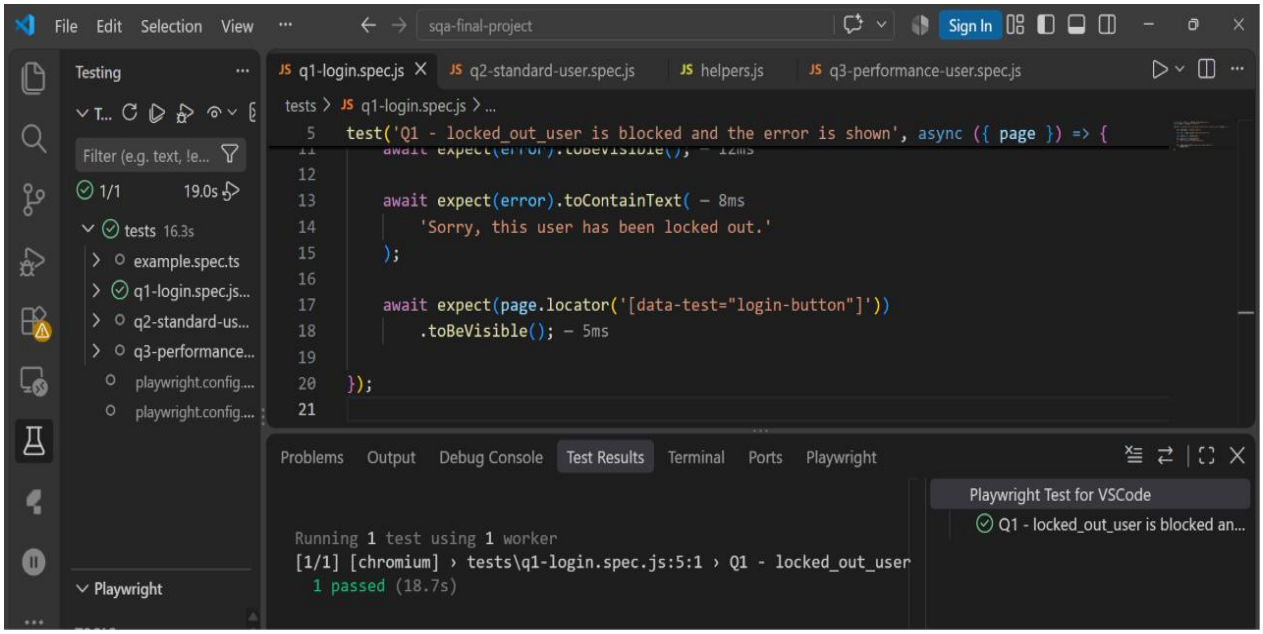
UI automation strategy

Each UI scenario is a separate test file. The shared helper module provides login, state reset/logout, checkout details, money parsing, and checkout-overview assertions. Tests use data-test attributes and Playwright locator assertions instead of arbitrary sleeps. The combined command supplies Q1, Q2, and Q3 files in that order.

Q1 – Try login in with locked_Out_User and Verify the Error Messge

Run Terminal: npx.cmd playwright test tests/q1-login.spec.js --project=chromium

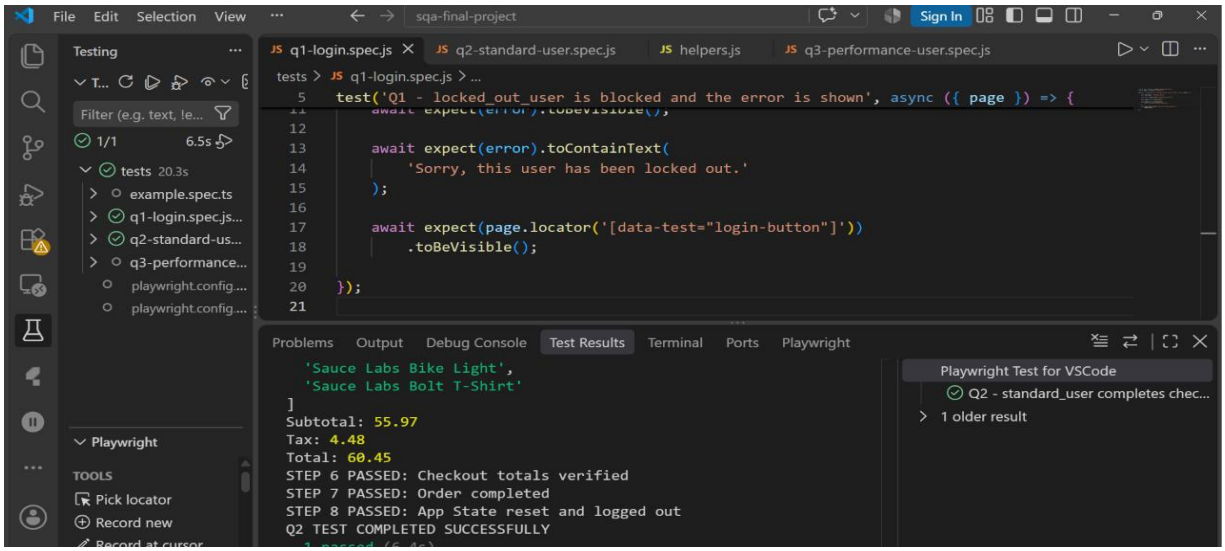
Steps: open SauceDemo, enter locked_out_user and the documented shared demo password, submit, assert the error is visible and contains the locked-out message



Q2 –

Run Terminal: npx.cmd playwright test tests/q2-standard-user.spec.js --project=chromium --headed --workers=1

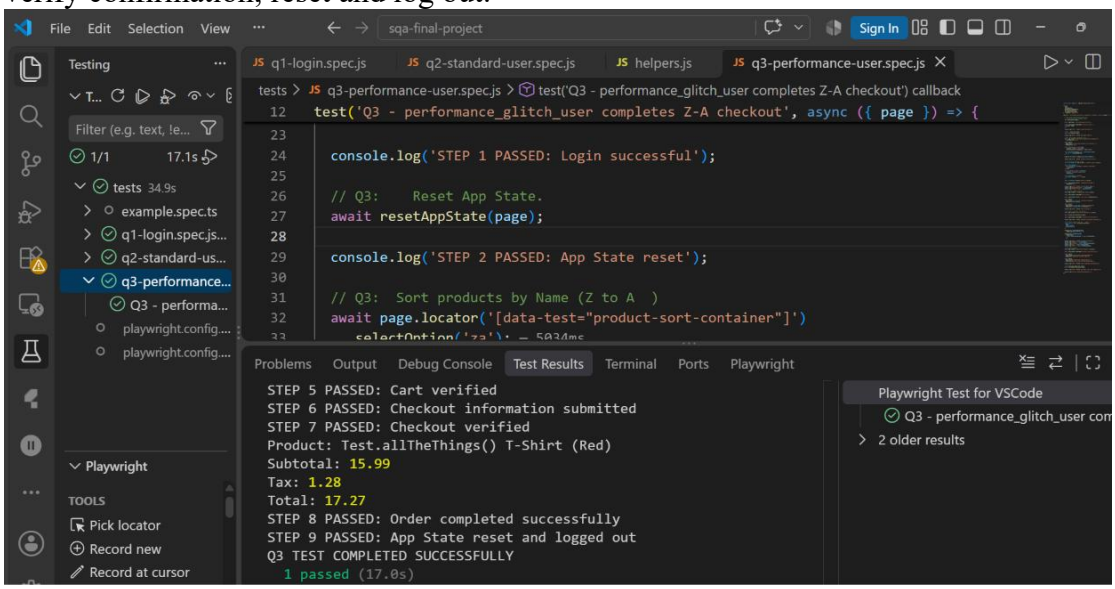
Steps: log in; reset app state; add first three inventory products; verify cart names and count; enter checkout information; compare overview names and displayed subtotal/tax/total; finish purchase; verify success message; reset app state and log out.



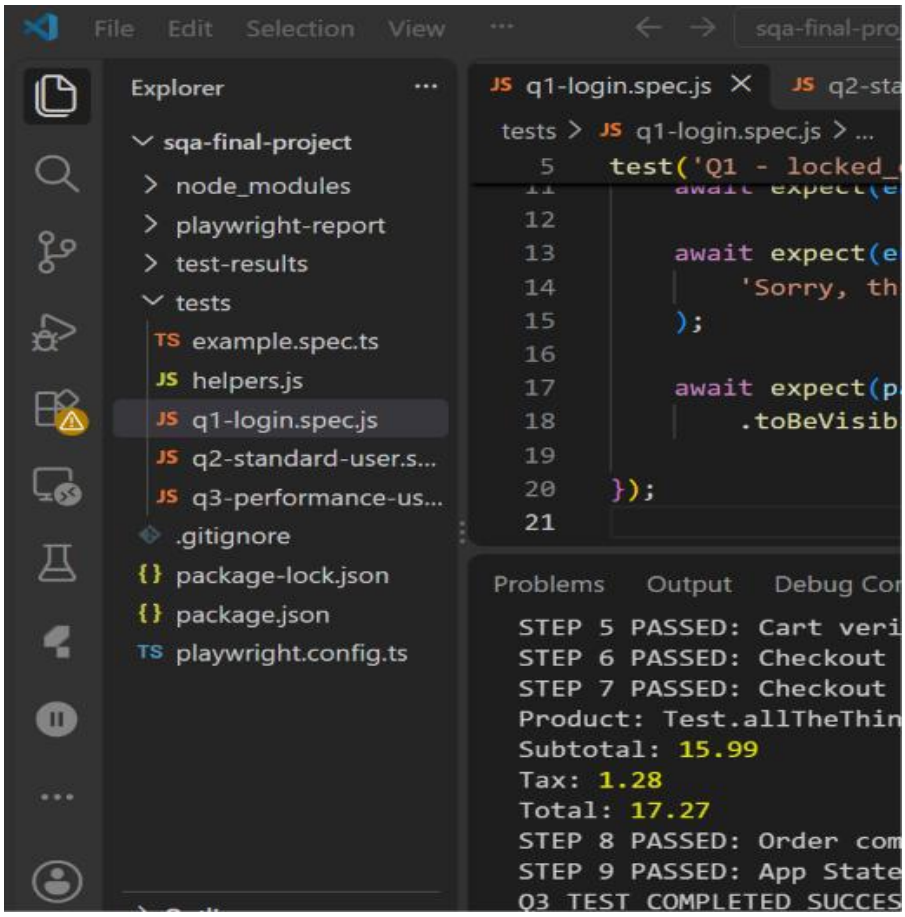
Q3 –

Run Terminal: npx.cmd playwright test tests/q3-performance-user.spec.js --project=chromium --headed --workers=1

Steps: log in; reset state; choose Z-to-A sorting; verify displayed names are descending; add the first product; verify cart and overview name/price totals; finish; verify confirmation; reset and log out.

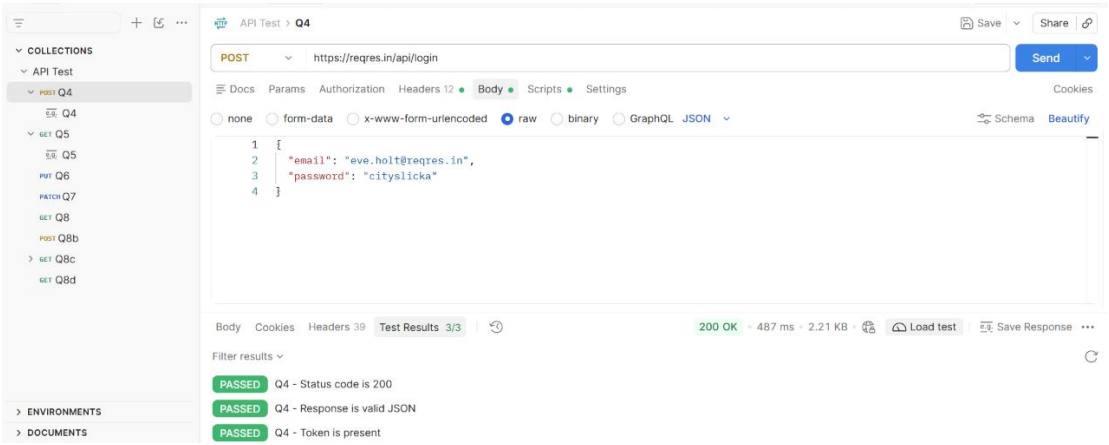


File:

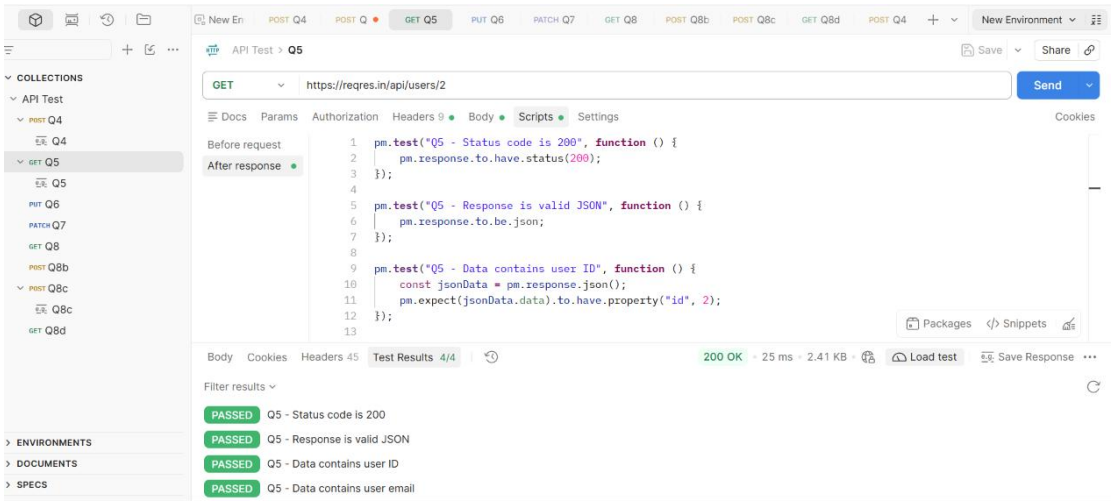


API automation

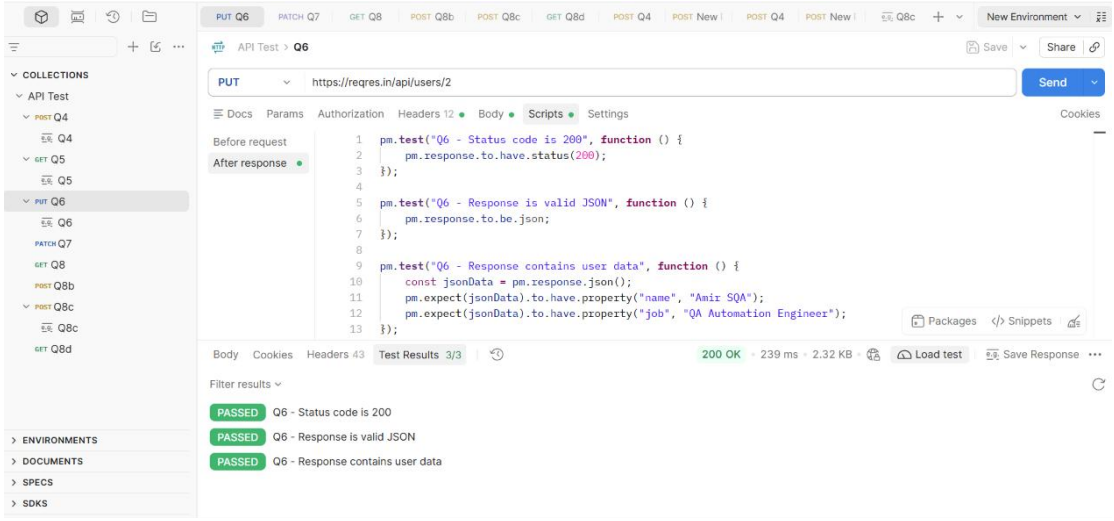
Q4 –



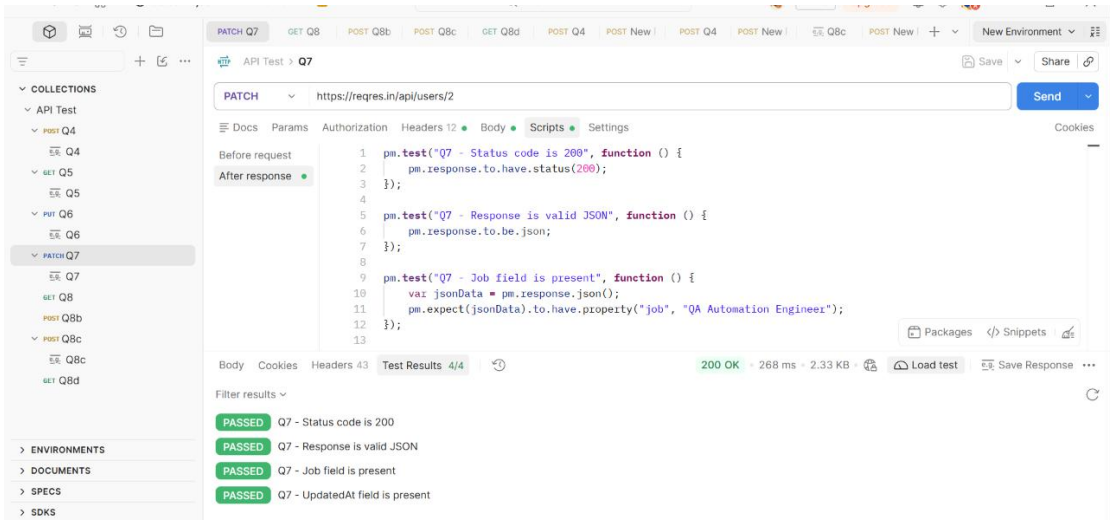
Q5 –


API Test > Q5
GET
https://reqres.in/api/users/2
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Before request
After response
pm.test("Q5 - Status code is 200", function () {
pm.response.to.have.status(200);
});
pm.test("Q5 - Response is valid JSON", function () {
pm.response.to.be.json;
});
pm.test("Q5 - Data contains user ID", function () {
const jsonData = pm.response.json();
pm.expect(jsonData.data).to.have.property("id", 2);
});
Body Cookies Headers 45 Test Results 4/4
200 OK 25 ms 2.41 KB
Filter results
PASSED Q5 - Status code is 200
PASSED Q5 - Response is valid JSON
PASSED Q5 - Data contains user ID
PASSED Q5 - Data contains user email

Q6 –

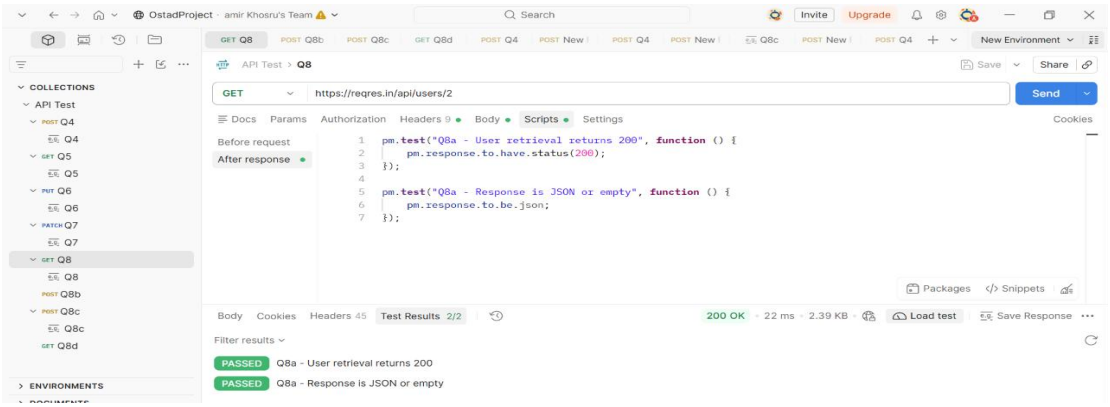

API Test > Q6
PUT
https://reqres.in/api/users/2
Send
Docs Params Authorization Headers 12 Body Scripts Settings
Before request
After response
pm.test("Q6 - Status code is 200", function () {
pm.response.to.have.status(200);
});
pm.test("Q6 - Response is valid JSON", function () {
pm.response.to.be.json;
});
pm.test("Q6 - Response contains user data", function () {
const jsonData = pm.response.json();
pm.expect(jsonData).to.have.property("name", "Amir SQA");
pm.expect(jsonData).to.have.property("job", "QA Automation Engineer");
});
Body Cookies Headers 43 Test Results 3/3
200 OK 239 ms 2.32 KB
Filter results
PASSED Q6 - Status code is 200
PASSED Q6 - Response is valid JSON
PASSED Q6 - Response contains user data

Q7 –

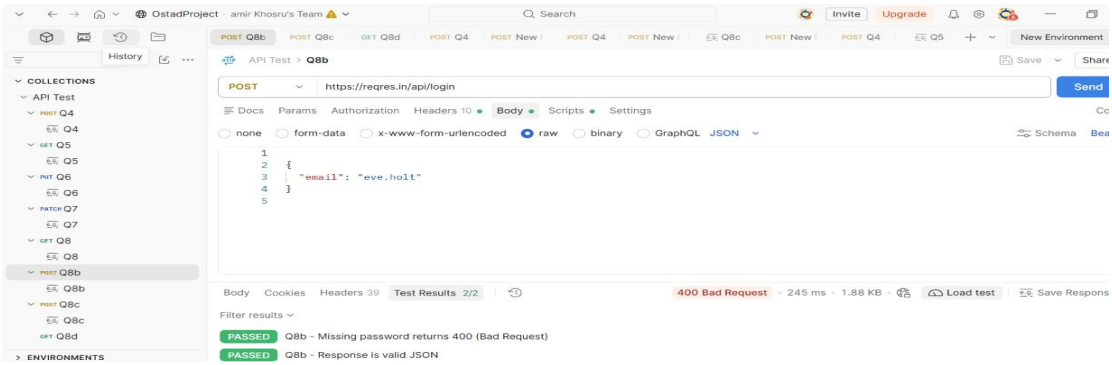

API Test > Q7
PATCH
https://reqres.in/api/users/2
Send
Docs Params Authorization Headers 12 Body Scripts Settings
Before request
After response
pm.test("Q7 - Status code is 200", function () {
pm.response.to.have.status(200);
});
pm.test("Q7 - Response is valid JSON", function () {
pm.response.to.be.json;
});
pm.test("Q7 - Job field is present", function () {
var jsonData = pm.response.json();
pm.expect(jsonData).to.have.property("job", "QA Automation Engineer");
});
Body Cookies Headers 43 Test Results 4/4
200 OK 268 ms 2.33 KB
Filter results
PASSED Q7 - Status code is 200
PASSED Q7 - Response is valid JSON
PASSED Q7 - Job field is present
PASSED Q7 - UpdatedAt field is present

Q8 –

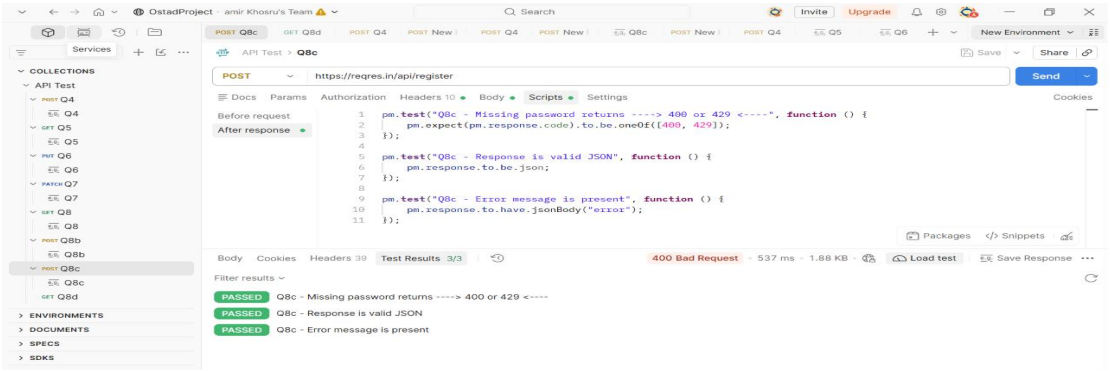
A


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Search
Invite
Upgrade
API Test > Q8
GET
https://reqres.in/api/users/2
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Before request
After response
pm.test("Q8a - User retrieval returns 200", function () {
pm.response.to.have.status(200);
});
pm.test("Q8a - Response is JSON or empty", function () {
pm.response.to.be.json;
});
Body Cookies Headers 45 Test Results 2/2
200 OK · 22 ms · 2.39 KB
Load test
Save Response
Filter results
PASSED Q8a - User retrieval returns 200
PASSED Q8a - Response is JSON or empty

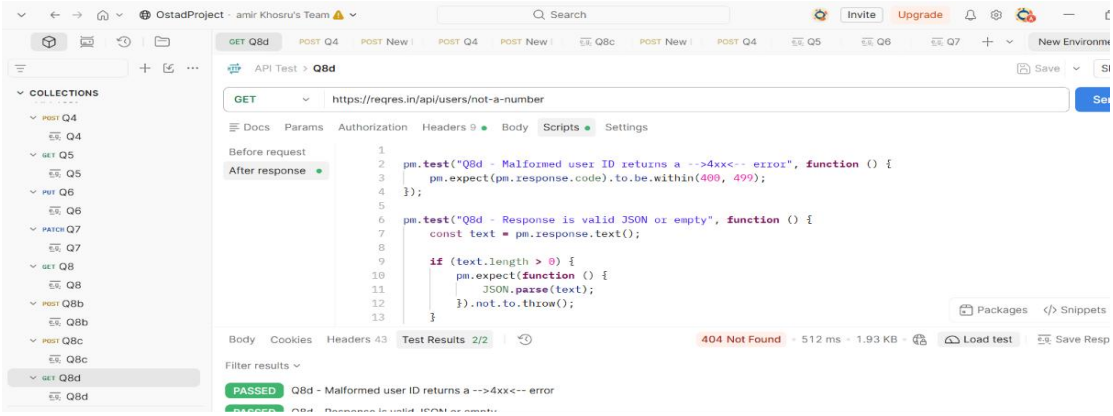
B


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Search
Invite
Upgrade
API Test > Q8b
POST
https://reqres.in/api/login
Send
Docs Params Authorization Headers 10 Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL JSON
{
"email": "eve.holt"
}
Body Cookies Headers 39 Test Results 2/2
400 Bad Request · 245 ms · 1.88 KB
Load test
Filter results
PASSED Q8b - Missing password returns 400 (Bad Request)
PASSED Q8b - Response is valid JSON

C


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Search
Invite
Upgrade
API Test > Q8c
POST
https://reqres.in/api/register
Send
Docs Params Authorization Headers 10 Body Scripts Settings
Before request
After response
pm.test("Q8c - Missing password returns ----> 400 or 429 <----", function () {
pm.expect(pm.response.code).to.be.oneOf([400, 429]);
});
pm.test("Q8c - Response is valid JSON", function () {
pm.response.to.be.json;
});
pm.test("Q8c - Error message is present", function () {
pm.response.to.have.jsonBody("error");
});
Body Cookies Headers 39 Test Results 3/3
400 Bad Request · 537 ms · 1.88 KB
Load test
Save Response
Filter results
PASSED Q8c - Missing password returns ----> 400 or 429 <----
PASSED Q8c - Response is valid JSON
PASSED Q8c - Error message is present

D


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Search
Invite
Upgrade
API Test > Q8d
GET
https://reqres.in/api/users/not-a-number
Docs Params Authorization Headers 9 Body Scripts Settings
Before request
After response
pm.test("Q8d - Malformed user ID returns a -->4xx<-- error", function () {
pm.expect(pm.response.code).to.be.within(400, 499);
});
pm.test("Q8d - Response is valid JSON or empty", function () {
const text = pm.response.text();
if (text.length > 0) {
pm.expect(function () {
JSON.parse(text);
}).not.to.throw();
}
Body Cookies Headers 43 Test Results 2/2
404 Not Found · 512 ms · 1.93 KB
Load test
Filter results
PASSED Q8d - Malformed user ID returns a -->4xx<-- error